

Integrated data structures in materials management as the foundation for circular economy in SMEs

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Abstract: SMEs are pivotal to Europe’s circular transition but often lack capabilities to acquire and operationalize sustainability data. The contribution proposes an integrated approach that anchors triple-bottom-line attributes as data fields in material and supplier masters and acquires them automatically in the background. A crawl–validate–integrate workflow harvests public sources, i.e. LCI/LCA repositories, EPDs, aligns records via ontology-based data access, and assures fitness with a machine-learning quality filter and human-in-the-loop curation. The minimal, extensible model includes among others material-level CO₂e, recycled content, recyclability rate and supplier location emissions. Provenance, method, and validity are centrally tracked. A first validation against SAP Materials Management indicates that a small set of Z-fields and a lightweight architecture (OBDA, REST/GraphQL, low-code ETL) can make indicators available to purchasing, planning or inventory management supporting decisions in operational business management and with enough data collected for strategical decisions on business models.

Keywords: SMEs, ERP systems, materials management, sustainability data, artificial intelligence, twin transformation, data-driven sustainability transformation, circular economy, digital product passport, automatic data acquisition, triple bottom line

1 Introduction

Since 2019, the European Green Deal (EGD) has defined the guideline for a climate-neutral European Union by 2050 and, with an interim target of a 55 percent reduction in greenhouse gases by 2030 [Th19]. It has established a legally binding transformation goal that explicitly focuses on resource efficiency and the circular economy. In all three core areas of the EGD – circular economy, energy efficiency and emission reduction the use of data is essential for identifying areas for action, monitoring, controlling and implementing actions to adress them [Th19]. The Circular Economy Action Plan (CEAP), which builds on this, specifies this requirement by highlighting digital product and process data as a key factor in making material flows transparent, closing resource loops and enabling new business models [Ci20].

Sustainability data according to CEAP can be defined as as all quantifiable economical

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cost and value as well as ecological and social impact factors of a product or process over its life cycle [Ci20]. This definition is related to the triple-bottom-line approach [Je00] which is one of the key concepts connecting economical performance with environmental and social impacts. This definition is being referred to throughout the whole contribution when the “sustainability data” is being used. On the one hand, it includes compliance data that is required by law and can be found, for example, in the mandatory fields of the European Sustainability Reporting Standards (ESRS) [Eu23], and, on the other hand, performative data that goes beyond the minimum legal requirements and contributes directly to value creation, for example in the form of real-time footprints or risk-based supplier metrics [KM25].

Small and medium-sized enterprises (SMEs) face structural limitations in view of this data paradigm because they often lack the financial and human resources to collect sustainability data in the required depth, harmonise it semantically and integrate it seamlessly into existing information systems [PM24]. The lack of standardised data structures in enterprise resource planning systems [W104] [AI24b] not only makes it difficult to fulfil growing reporting obligations in a cost-efficient manner, but also prevents the use of the performative potential of circular business models, threatening to create an uneven playing field between large companies and SMEs [Pü24].

Against this backdrop, this paper examines how sustainability data can be automatically embedded in material and supplier master and transaction data so that SMEs can make both operational decisions in procurement and scheduling and strategic decisions towards circular value creation on a valid, integrative data basis.

2 State of research

2.1 Drivers for collection of sustainability data

Drivers that motivate companies to collect and use sustainability data started shifting recently [Pü24] from the rule-driven understanding that prevailed. The focus was on complying with legal requirements such as the reporting and due diligence obligations of the Corporate Sustainability Reporting Directive (CSRD) [Co22], and on maintaining legally compliant emissions and supply chain metrics through mechanisms such as the Carbon Border Adjustment Mechanism (CBAM) [Ca23] following the logic of the “licence to operate” paradigm. However, it this compliance is costly and often overwhelming for SMEs [PM24]. Recent political developments confirm this assessment: The EU’s so-called Omnibus Initiative has reduced the scope of disclosure originally expected by the European Sustainability Reporting Standard by 68% of mandatory disclosure points in the short term and eliminated mandatory disclosure of listed SMEs starting from 2027 as SMEs in particular would otherwise have been unable to cope with the effort [EU25].

Parallel to this debate on relief, a survey of 1,600 corporate and financial executives conducted by the World Economic Forum (WEF) points to the growing strategic importance of integrated sustainability and financial data as performative data. The vast majority of executives asked see the linking of both data worlds as a substantial competitive advantage that goes far beyond mere compliance effects [KM25]. The survey emphasises that reliable Environmental, Social, Governance (ESG) indicators are now perceived as "strategic asset for agility and innovation" and are increasingly weighing as heavily as traditional cost factors such as raw materials or energy in investment and pricing decisions [KM25].

Sustainability data is no longer solely assigned to reporting systems, but identified as an element of performative corporate management that must be embedded in operational decisions if efficiencies are to be realised and new business models developed [PM24]. Integrated data structures that combine financial and sustainability-related information and can be implemented technically without prohibitive costs are considered a decisive lever in making this change accessible to SMEs as well. Only the combination of real-time production and supplier data with environmental and social indicators paves the way for data-driven optimisation and innovation processes [Pü24]. The semantic integration hurdles that arise specifically in heterogeneous IT landscapes, underscore the need for domain-specific data models and lightweight interfaces [PM24].

2.2 Processing of sustainability data in Enterprise resource planing software

Conventional enterprise resource planning (ERP) systems have been the basis of the IT infrastructure of many industrial companies for decades, but their primary strength lies in the procedural integration of finance, logistics and production, not in processing and analytics of sustainability information [A125]. A case study on SAP S/4HANA shows that environmental indicators such as recycling rates cannot be recorded or evaluated in the standard model. Only after expanding the material master tables and adding additional business add-ins such indicators can be mapped at all [AG18]. Without expanding the material master, only a very limited ecological scope is available, making further analyses of product sustainability virtually impossible

This structural gap is particularly serious because ERP systems are widely used in the relevant industries: 95 percent of large manufacturing companies [Stat17] and 31% of all companies used in Germany an ERP solution [An24]. But even widely used products have so far only considered energy management to be a "marginal issue," so that despite high ERP penetration, there are hardly any integrated functions for environmental or resource data.

In the scientific debate, this gap is increasingly being described as an opportunity for innovation. Wunderlich et al. diagnose a clear "fit/gap" between the data requirements of a triple bottom line approach and the existing ERP data objects and call for domain-specific extensions and ontology tools to bring together material and sustainability data

[Wu18]. Putrama and Martinek classify these integration hurdles as typical for heterogeneous SME landscapes and see the core of the problem less in missing report exports than in a lack of semantic consistency, which prevents advanced analyses such as real-time CO₂ calculations [PM24]. Conceptual approaches to sustainable ERP, which since Chofreh et al. have been promoting an expanded range of functions along the three dimensions of sustainability, confirm the feasibility of such extensions, but point to the considerable investments in data architecture, project management and strategic anchoring that only large companies have been able to afford so far [CG14].

2.3 Standardized data exchange with the Digital Product Passport (DPP)

The DPP is regarded as a key instrument for the technical implementation of the transparency and circularity targets defined in the Ecodesign Regulation for Sustainable Products (ESPR). The latest CEN-CENELEC Workshop Agreement (CWA 18186:2025) explicitly refers to the DPP as an integrated information tool for the circular economy because it provides, for the first time, a structured, digitally accessible data core for all actors in a value chain [Gu25]. The European Commission expects that this will support data-based purchasing, usage and end-of-life decisions, thereby closing material cycles [Re24].

The EU project CIRPASS in particular clarifies in its system roadmap report from 2023, that the necessary DPP standards should be finalised "within the next five years" and that the first interoperable implementations should be available by 2027 for the priority product groups of batteries, textiles and electronics [Jo24].

While the data structures for most product groups are still pending, DIN SPEC 99100 provides a complete schema for the battery passport, which defines 130 mandatory attributes and a unique QR code and will be mandatory under the Battery Ordinance from 2027 [DI25]. But this content-fixed model will be of limited use without consistent semantic mappings in existing ERP environments because heterogeneous identifier systems, proprietary data formats and missing ontologies hinder automated data exchange [PM24]

The recently adopted European Data Act (EDA) can be seen as an catalyst, as it legally secures fair access to IoT and product data from 2025 onwards, thus creating the basis for open, cloud-based DPP platforms [Eu23b]. However SMEs, given their limited financial and IT resources, will hardly be in a position to implement complete DPP workflows and establish the necessary semantic data structures in the short term [Jo24] [PM24]. As SMEs account for the vast majority of companies, and a big share of gross domestic product in Germany [An25] [Ve25] but also other EU countries like Italy or Spain [Ma24] such a delay would significantly impair the ecologically and economically desirable broad impact of the circular economy.

2.4 Integration of heterogeneous sustainability data in business software

Traditional Extract-Transform-Load (ETL) processes, in which data is physically extracted, transformed and loaded into dedicated repositories, are considered costly and inflexible because each new data source requires extensive mapping and loading procedures [PM24]. As a resource-saving alternative virtual integration approaches in which federated queries or so-called data fabrics leave the data at the source and only link it at runtime can be used. A promising methodology is Ontology-Based Data Access (OBDA), which maps relational tables using a semantic intermediate schema, thus enabling technical queries across heterogeneous structures [Ca19]. Virtual integration patterns could serve as a resource-saving alternative that keep data at the source and link it only at query time, thus reducing duplication and improving timeliness [PM24].[Ca19]

Engines such as Ontop have matured to support SPARQL-to-SQL rewriting with soundness guarantees and industrial-scale performance [Ca19]. Application ontologies can act as pragmatic gateways into OBDA setups in manufacturing “Internet of Production” contexts, lowering adoption barriers by aligning operational schemas to domain vocabularies incrementally [Br23]. Nevertheless, even modular OBDA setups depend on sensible ontology scope and careful mapping governance; developing and maintaining such domain artefacts still requires expertise and time [PM24]

Complementary to semantics, machine learning (ML) is increasingly leveraged to reduce manual data wrangling. “Humans-in-the-loop” AutoML configurations are emphasised for schema/signal discovery under governance constraints, showing that active learning and curated feedback substantially improve label quality and mapping suggestions in real settings [CF21] [Br23].

From a data-engineering angle, automated feature extraction, anomaly detection and outlier handling are considered as critical building blocks in automated data processing pipelines, where data quality assurance is a prerequisite for trustworthy downstream analytics [MM25] [Br23]. Earlier semantic-integration studies report progress on (semi-)automatic mapping between ontologies and relational schemas which are useful for aligning sustainability attributes to ERP tables. But noisy or sparse input can degrade mapping fidelity without manual curation checkpoints [AG18] [WW17]

Overall virtual, semantics-based and ML-assisted integration promises considerable rationalisation potential for sustainability data, but their effectiveness depends significantly on the availability of consistent ontologies, meaningful training data and established data quality metrics.. SMEs in particular face the challenge of creating these infrastructural prerequisites with limited resources and no proficiency, making the development of lightweight, preconfigured integration solutions a priority research area [Br23] [Ca19] [PM24].

3 Need for action

3.1 Extending operational data structures in ERP systems

Materials management in enterprise systems rests on a small set of stable artefacts, most prominently the material master and the supplier master, around which operational functions such as purchasing, inventory management and inventory analysis are organised [Go22]. The material master provides the canonical key for planning, sourcing and valuation (for example item identifiers, units of measure, price or standard cost, lead times and bill-of-materials linkages), while the supplier master anchors commercial terms, locations and compliance status [Go22]. In current practice, these masters are optimised for economic control. Environmental and social information even when present, is often handled in separate tools or spreadsheets, which breaks traceability across transactions and impedes systematic evaluation of alternatives in day-to-day operations [WW17] [FJ24]. Isolating environmental and social indicators outside the operational data backbone leads to manual joins, media breaks and poor reuse in processes such as purchasing, scheduling and inventory analysis [Br23]. Therefore extending core masters with sustainability fields is a prerequisite for meaningful life-cycle-based assessments and automation along the materials flow [AG18] [Go22]

System analyses of ERP-systems like SAP S/4HANA show that the lack of corresponding attributes blocks the derivation of holistic life cycle assessments[Wu18]. Without fields for CO₂ factors, recycled content, durability/repair proxies or hazardous-substance flags in material masters and transaction records, life-cycle assessments, take-back logic, and reuse/remanufacturing planning remain manual, error-prone and fragmented [Ve25].

The expansion of master data by fields for sustainability data represents the lowest common denominator for covering both regulatory reporting requirements and performative analyses. This means that in future, each material object could, for example, be assigned the recycled content, a CO₂ equivalent and a hazard indicator in addition to quantity, quality and value variables. Supplier entries could be supplemented by, i.e. location emission factors and social indicators supporting supply chain due diligence. Latter will need further research and will be discussed in a future contribution.

CIRPASS already implements the attribute logic for the Digital Product Passport by requiring the consistent assignment of emissions, circularity and quality-related data along the value chain [Jo24]. Once the data fields are integrated in the master data of ERP-systems, automated assignment procedures can pull and update values from open sources such as the Oekobaudat database, Ecoinvent database or future DPP registers, significantly reducing the cost and personnel for SMEs.

Tab 1. illustrates extended data fields based on regulatory requirements; it shows that relevant information can largely be derived from existing open data sources or future DPP registers, provided that the master data has the necessary semantic capacity.

Master data object	Example KPI	Public data source	Regulatory requirement
Material	CO ₂ equivalent per unit	Ecoinvent	CBAM
	Recycled Content (%)	DPP-Register	ESPR
	Risk Index	ECHA-SVHC	REACH
Supplier	Location emission factor	EDGAR	CBAM
	Social risk score	UNDP-Atlas	CSDDD
	Certificate status	ISO 14001	CSR

Tab. 1: Data object with sustainability attribute, public data source and regulatory requirements

3.2 A pragmatic approach to data collection

The state of research illustrates that classic ETL routines are neither viable for SMEs from considering financial and personal resources. Virtual, ontology-based integration methods only deliver reliable results if the underlying raw data is continuously updated and quality-assured [PM24]. This leads to the need for a pragmatic data collection approach, that relies on automation, open data sources and machine learning plausibility checks.

Automated crawlers can continuously extract data from public sources such as Ecoinvent and other public data bases (see Tab. 1) or pre-stage data already available from the DPP sandbox at specified intervals. This would follow the recommendation of the CIRPASS roadmap to use open product and process data as a low-threshold entry point into the DPP logic [Jo24]. Access is legally supported by the interoperability requirements of the Data Act, reducing licensing barriers to a minimum [Eu23b].

Another important technique is semantic matching. ODBA methods are used to match the heterogeneously structured crawler results with the additional data fields of the material and supplier master data. This enables consistent and accurate assignment of data to the corresponding attributes in the master data.

An AI-supported quality module addresses the data quality risk that has been repeatedly criticised in the literature. An active learning ensemble assigns a confidence value to each data record and only forwards entries with a confidence value below i.e. 0.8 to human reviewers. This significantly reduces the manual validation effort, as only exceptional cases require personal expertise, while routine updates are fully automated. [WW17].

Such a crawl–validate–integrate workflow could close the gap between existing open data sets and the extended ERP master data. At the same time, it meets the two key requirements highlighted by the SME survey in CIRPASS: minimal staffing levels and

immediate system compatibility [AI24a]. This creates a robust data basis that enables both reporting on environmental impacts, i.e. Scope 3-CO₂-emissions, required by regulations and the performative use of sustainability indicators in day-to-day business.

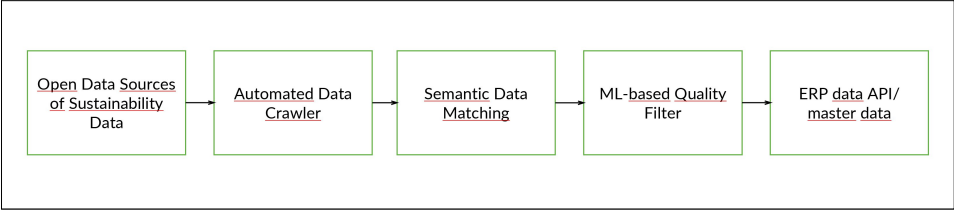


Fig.1 Crawl-validate-integrate pipeline

Figure 1 outlines the resulting crawl-validate-integrate pipeline and illustrates how the AI-supported quality filter could be inserted between data extraction and ERP persistence. By combining these approaches, SMEs could efficiently and cost-effectively collect and use their sustainability data to make informed decisions and improve their sustainability performance.

3.3 Closing the architectural gap for extended data structures in SME software

Missing sustainability fields in ERP systems [Wu18], cost-driving ETL pipelines [PM24] and DPP interfaces that will only be fragmentarily standardised by 2027 [Jo24], identified as deficits of ERP-systems (see Tab. 2), make it clear that SMEs need a new type of software architecture that extends existing business applications without requiring extensive core adjustments. Research and project experience point to a hybrid integration paradigm that combines three elements: an ontology layer for semantic interoperability, lightweight REST or GraphQL APIs for real-time data exchange, and low-code tools to automate routine ETL tasks.

Ontology-based data access allows domain-specific sustainability terms to be mapped to relational material and supplier tables without forcing physical data replication; OBDA approaches can reduce implementation effort by up to forty percent compared to classic data marts, if the ontology is encapsulated as a separate service layer [PM24]. The CIRPASS roadmap for the DPP recommends a decentralised resolver concept based on REST and GraphQL endpoints to make product and passport information accessible beyond proprietary ERP boundaries [Jo24]. Thirdly, practical reports on cloud-based environmental information systems show that low-code ETL engines can reduce the ongoing maintenance effort for period runs even in SMEs if they are deployed on container-based open source stacks [WW17].

Current capabilities of typical SME ERP systems	Missing function block	Recommended technology component
Static material and supplier master data without sustainability attributes	Semantic mapping of CO ₂ , risk and recycling data	Ontology layer (OBDA)
Point-to-point interfaces, proprietary formats	Open, transaction-oriented data access	REST/GraphQL-API-Gateway
Manual CSV uploads for CO ₂ factors	Ongoing, automated data integration	Low-Code-ETL with Container-Deployment

Tab. 2 Architectural shortcomings of today's ERP landscapes and recommended technology layers based on Wu18, WW17, PM24, CI24

Closing this architecture gap is a prerequisite for technically supporting the mandatory field logic formulated in section 3.1 and for permanently operating the crawl–validate–integrate pipeline outlined in section 3.2. The specific design of such an extension package is presented in the following chapter.

4 Proposed concept

The proposed solution is aimed at SMEs. As these enterprises lack the financial [Du25] and personal resources for collecting sustainability data as well as the required proficiency for leveraging the harvested data in their business operation this contribution proposes an integrated data structure where sustainability data is integrated into material management data and collected automatically in the background.

4.1 Integrated data structures in materials management

Foundational for an integrated data structure are three requirement classes: (i) regulatory requirements from current and upcoming EU or national legislation (e.g. CBAM, ESPR), (ii) performative requirements to enable analytics that improve enterprise performance, and (iii) functional requirements in purchasing, inventory analysis, and inventory management.

From a regulatory perspective, several EU instruments now explicitly require product- and supplier-related sustainability information in machine-readable form. CBAM mandates declarations of embedded emissions for specified goods and distinguishes verified actuals from default values with verifier-backed records [Ca23]. ESPR and the forthcoming DPP standardise information sets for carbon, resource use, substances of concern, and circularity characteristics (e.g. durability, repairability, recycled content, recyclability), with links to ISO 14044/14067, the GHG Product Standard [Gree11], and emerging circularity metrics [Is25]. CWA 18186 specifies the corresponding data structures and exchange semantics for product and component data [Gu25]. The Battery

Regulation already mandates a QR-based passport with recycled-content targets and operational traceability from 2027, coupling compliance with circular outcomes such as higher secondary-material shares and second-life design [Co22b]. Meeting these obligations without media breaks requires that recycling indicators and product/supplier-level emissions be available as structured master-data attributes [DI25].

Anchoring triple-bottom-line (TBL) data—environmental, social, and economic—as first-class fields in material and supplier masters is therefore essential, especially for SMEs. It creates a shared “sentence structure” for analytics and interoperability [AG18][PM24][Gr14], enabling consistent capture of product- and supplier-level impacts (including Scope-3 CO₂ emissions) across life-cycle stages and their translation into operational and strategic decisions such as secondary-material sourcing, repair, reuse, and remanufacturing. Non-financial indicators (e.g. product-level carbon intensity, recycled content, due-diligence status) must be structured and linked to financial performance via cause-and-effect chains to guide resource allocation rather than merely external reporting [Fi01][Ca19]. Integrating environmental metrics into planning, budgeting, and variance analysis—emphasising data quality, targets, and threshold-based steering—requires stable, queryable attributes in operational masters instead of ad-hoc spreadsheets [Gr14][KM25].

Functionally, the integrated data structure serves three purposes. Purchasing, inventory management and inventory analysis.

In purchasing, sustainability attributes must be queryable alongside price and lead time to rank suppliers and items by combined economic-environmental-social KPIs (e.g. EUR/ton, kg CO₂e/ton, social-risk index) and to support CBAM-ready documentation at goods-receipt and invoice-line level [Ca23][Gu25]. In inventory management, master-level circularity fields—recycled content, repairability, remanufacturability, recyclability—support segregation strategies, substitution rules, and take-back allocation; ESPR/DPP guidance makes these attributes machine-resolvable for decisions across hand-over points [Gu25][Re24]. In inventory analysis and planning, life-cycle and Scope-3 factors should join the same fact structures as quantities and values to permit period close, scenario analysis, and supplier switching with sustainability deltas computed like standard variances; environmental informatics and eco-controlling literature recommends embedding such indicators in the enterprise backbone to enable alerts and thresholds in operational control loops [WW17][AG18].

To satisfy these regulatory, performative, and functional requirements—and to prepare for the DPP—the data model links the material master to a minimal, extensible set of TBL attributes, complemented by supplier parameters for location-based CO₂ emissions and due-diligence status. Applying recognised standards ensures methodological consistency and comparability: ISO 59020 and DIN SPEC 99100 for circularity-related data, ISO 14067 and the GHG Product Standard for carbon measurement [Is25][DI25][Th19]. Storing recycled content, resource-use indicators, and durability/repair proxies at master

level makes them available to procurement, MRP, and recovery processes without custom joins and enables after-product-life KPIs (e.g. reuse, recyclability) in standard queries and dashboards [Gu25].

For social information, supply-chain due-diligence and certification status should be captured in supplier masters to enable risk-aware sourcing and audit planning; role-based access and long-term availability are addressed explicitly in the DPP guidance [Gu25]. Finally, forward compatibility with the DPP requires stable identifiers (e.g. GS1/QR links), versioning, and resolvable links to external registries so that machine-readable passports can be connected as delegated acts roll out and as JTC-24 APIs for searchability and life-cycle management become available [Gu25].

Data field name	Data field Description	Typical source
Dangerous-goods profile	Standard ERP profile for DG handling	ERP
Product carbon footprint	Cradle-to-gate CO ₂ e per declared unit (CBAM-ready)	EPD/LCI (ISO 14067; 14044/EN 15804); Supplier
Recycled content (%)	Mass-based recycled share	EPD/DPP (ISO 59020; CWA 18186); Supplier
Recycling rate at EoL (%)	Mass fraction entering recycling at end-of-life	ISO 59020; DPP/CWA 18186
Repairability index	Proxy for ease of repair/upgrade (0–10 or scheme)	DPP/CWA 18186; ISO 59020
Durability class	Proxy for expected lifetime category	DPP/CWA 18186; ISO 59020
Primary energy demand	Total primary energy per declared unit	EPD/LCI (ISO 14044; EN 15804; CWA 18186)
GS1 GTIN	Global Trade Item Number for product identification	GS1 (forward compatibility for DPP)
DPP UID	Digital Product Passport unique identifier	DPP registry (CWA 18186; CEN/CENELEC JTC 24)
Supplier Due-diligence status	Supply-chain due diligence alignment	Supplier/Compliance (CSDDD)
Geography scope	Geographical scope of dataset	Source (CWA 18186 info sets)

Tab.3: Excerpt from the integrated data structure for materials management

Tab. 3 is showing an excerpt of the proposed data structure including the typical sources². The full data structure is consisting of 40+ additional data fields that are integrated with existing, industry specific materials and supplier management master and transaction data structures.

4.2 Acquiring and processing heterogeneous sustainability data from public sources

As sustainability transformation is no exact science, publicly accessible sustainability data constitute a pragmatic foundation for building sustainability management capacities in the majority of SMEs. Therefore secondary datasets from public or private sources of life-cycle inventory (LCI) and assessment (LCA) data as well as environmental product declaration (EPD) (see Tab. 4) are considered sufficiently robust to provide a baseline and to track directional improvements over time. Current integration research recommends leveraging existing open or low-barrier sources first and increasing primary data share progressively as standards such as the Digital Product Passport (DPP) mature [PM24][CI24]. In practice, this implies acquiring and validating from two classes of sources: I) structured, machine-readable endpoints (APIs, CSV/RDF feeds) and ii) semi-structured documents (typically PDF-based EPDs or producer datasheets) that require controlled extraction and semantic anchoring.

Public data source	Content coverage	Update interval	Licence agreement
Ecoinvent v3.10	Life cycle CO2 factors for > 20,000 materials	annually	CC-BY-NC
EDGAR v8.0	Territorial emission factors (Scope 3)	half-yearly	CC-BY-4.0
EU DPP Sandbox	Preliminary data on batteries, textiles, electronics	quarterly	EUPL-1.2
ECHA-SVHC	Hazardous substance classification	continuously	CC-BY-4.0
UNDP-Atlas	Social risk index (country level)	annually	CC-BY-NC

Tab.4: Public available Data sources with content coverage, update intervall and license

An important requirement for integrating the heterogeneous sustainability data from two classes of sources are a robust acquisition routine and integration workflow (see Fig. 2). Given the limitations of SMEs in terms of the high cost of physical ETL processes and the expertise required for robust semantics, this contribution proposes a

² Final structure still subject to development. Release planned for 12/25.

crawl-validate-integrate workflow that combines automation, open data sources, and an ODBA- and ML-based plausibility checks to populate the extended material and supplier data.

Low-overhead crawlers harvest public data relevant for material and supplier attributes (see Tab. 3), prioritising authoritative sources and structured endpoints where available. For structured endpoints, like Ecoinvent, background services perform scheduled retrievals that expose material- or process-level indicators (for example greenhouse-gas intensities, recycled content shares, or hazard classifications). The capture logic prioritises documented interfaces and machine-readable formats in order to minimise the effort required for downstream data processing and to guarantee the origin of the data. This strategy is advocating open product and process data, interoperable interfaces and decentralised resolvers as emphasized by CIRPASS [CI24]. The European Data Act can further reduce legal and technical barriers for programmatic access and switching between processing services from September 2025 onward, thereby strengthening SMEs' ability to collect and reuse device- and service-generated data under interoperable conditions [EU23b].

The second class of sources are semi-structured documents, which demand a different treatment. Many producers publish EPDs and sustainability datasheets as downloadable PDFs; these artefacts often encode declared units, system boundaries and impact modules in human-readable layouts rather than in clean tables. The acquisition routine therefore combines template-guided parsing with layout-aware extraction and applies explicit unit normalisation, code-list checks and completeness tests before accepting any values.

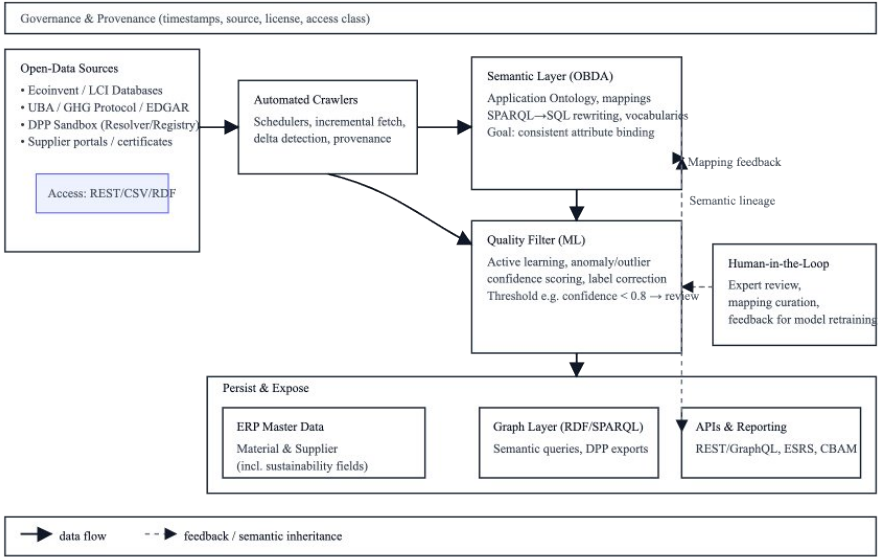


Fig. 2: Automated Crawl-Validate-Integrate Workflow with Human-in-the-Loop feedback

Where templates vary substantially across issuers or product groups, a semantic anchoring step is required. The proposed solution will use application ontologies that provides a controlled vocabulary for key attributes (for example “global warming potential per declared unit,” “recycled content share,” “hazard class”), while OBDA mappings bind extracted fields to this vocabulary without forcing physical replication of the source data [Ca19]. Manufacturing case work demonstrates how such application ontologies can act as gateways into OBDA setups and thus stabilise matching logic across heterogeneous artefacts [Ba23]. This semantic step is essential to ensure that values originating from distinct layouts or issuers become comparable, searchable and auditable.

To mitigate the well-documented data-quality risks in sustainability datasets, the pipeline inserts an ML-based quality filter between ingestion and ERP persistence. Following enterprise AutoML practice, a human-in-the-loop scheme flags low-confidence assignments (e.g., confidence < 0.8) for expert review, while high-confidence cases are auto-accepted; this reduces manual effort without sacrificing accountability [FG21] [MM25]. For noisy or weakly labelled inputs—typical in scraped PDFs—modern label-quality methods can identify and correct suspect labels during training, which measurably reduces false positives in subsequent anomaly detection [SYM23].

The crawl–validate–integrate pattern closes the gap between open datasets (and future DPP registries) and extended ERP master data. It aligns with CIRPASS’ interoperability stance and the Data Act’s interoperability objectives, while matching SME realities [CI24] [EU23b]

5 Discussion

The proposed crawl–validate–integrate approach addresses key concerns in environmental informatics: material efficiency through data-based transparency and the use of open data sources. The lightweight architecture based on open-source software services (OBDA layer, API gateway, low-code ETL) can be placed in front of business software or ERP cores, minimising customisation effort. As machine learning can support the validate-phase and experiences in SMEs shows adaptable ML methods can improve efficiency and reduce waste rates [WWR22][GBB21], yet consistent schemas and domain knowledge remain limiting factors. Research needs to clarify which thresholds (e.g., confidence 0.8) are optimal, how much maintenance time active learning saves in real data flows, and how robust ontology-based matching (OBDA) remains against noisy open data feeds [SYM23].

Also how stable semantic links actually are across heterogeneous keys (material numbers, certificate IDs) and how governance/provenance is operationalised (e.g. change/audit trails) needs to be clarified.

The need for research should be addressed in lighthouse projects or reference implementations with metrics (integration effort, error rate, throughput time) and comparison to physical ETL alternatives, also in cooperation with ERP manufacturers.

Finally, the environmental impact of digital infrastructures themselves requires assessment. Always-on pipelines and cloud services consume significant amounts of energy and resources, and rebound effects can offset efficiency gains. Studies on ICT-related raw materials highlight risks across the life cycle [WOF23][WWR22]. Research should determine the net ecological balance of SME-scale crawl–validate–integrate pipelines, exploring operating modes (e.g., batch frequency, edge vs. cloud) that minimise footprints while maintaining data quality.

The proposed concept envisages a decoupled architecture: lightweight, preferably open-source services (OBDA layer, API gateway, low-code ETL) in front of the ERP core to ensure release capability and gradual introduction. This encapsulation takes into account typical SME constraints and minimises deep core customisation. It remains unclear how stable semantic links across heterogeneous keys (material numbers, certificate IDs) actually are and how governance/provenance is operationalised (e.g. change/audit trails). The need for research should be addressed in lighthouse projects or reference implementations with metrics (integration effort, error rate, throughput time) and in comparison to physical ETL alternatives, also in cooperation with ERP manufacturers.

Finally, the environmental impact of digital infrastructures themselves requires assessment. Always-on pipelines and cloud services consume energy and resources, and rebound effects can offset efficiency gains. Studies on ICT-related raw materials highlight risks across the life cycle [WOF23][WWR22]. Research should determine the net ecological balance of SME-scale crawl–validate–integrate pipelines, exploring operating

modes (e.g., batch frequency, edge vs. cloud) that minimise footprints while maintaining data quality.

6 Conclusion

Empowering SMEs for the twin transformation of digitalisation and sustainability requires an integrated approach. On one hand, digital maturity must increase significantly; on the other, ecological and social data should be collected with minimal additional effort and embedded directly in operational decision-making in order to support the adaption of sustainable, circular business models in the medium to long term. Because SME management is strongly economically oriented, adoption is most likely when clear business cases show disproportionate cost savings or revenue potential. Open questions remain: What economic arguments actually motivate SMEs to digitise and consider sustainability data in their decisions?

The contribution shows that integrated data structures along the triple bottom line, anchored as attributes in the material and supplier master, can form the basis for making sustainability information operationally usable in SMEs. In conjunction with the proposed crawl–validate–integrate pipeline, which combines open data sources, semantic access (OBDA) and an AI-supported quality filter, an implementation path is outlined that addresses regulatory requirements (CBAM/ESPR/DPP) and performative control needs in day-to-day business. The data set and processing architecture can be considered as a foundation for integrating circularity-relevant key figures into procurement, scheduling, inventory analysis and return processes, thereby enabling data-based cycle orientation in regular operations.

Open issues regarding scalability and robustness of semantic mappings across heterogeneous data spaces, the net effort of continuous data collection and maintenance (including human-in-the-loop components), and the net ecological impact of the digital infrastructure itself should be addressed. In a field test or a lighthouse project, e.g. with an ERP manufacturer reliable evidence of successful implementation should be created. In a representative product group the effectiveness of the approach should be measured using defined metrics such as allocation accuracy, data maintenance effort, savings and substitution potential in procurement, and material and energy consumption, including possible rebound effects.

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